

## 1. Read and display a 2d array

```
Import java.util.Scanner;
```

```
Public class ReadDisplay2D {
```

```
    Public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter rows: ");
```

```
        Int rows = sc.nextInt();
```

```
        System.out.print("Enter columns: ");
```

```
        Int cols = sc.nextInt();
```

```
        Int[][] arr = new int[rows][cols];
```

```
        System.out.println("Enter elements:");
```

```
        For (int i = 0; i < rows; i++) {
```

```
            For (int j = 0; j < cols; j++) {
```

```
                Arr[i][j] = sc.nextInt();
```

```
            }
```

```
        }
```

```
        System.out.println("2D Array:");
```

```
        For (int i = 0; i < rows; i++) {
```

```

        For (int j = 0; j < cols; j++) {
            System.out.print(arr[i][j] + " ");
        }
        System.out.println();
    }
}
}

```

2. Find the sum of all element

Import java.util.Scanner;

Public class Sum2DArray {

Public static void main(String[] args) {

Scanner sc = new Scanner(System.in);

System.out.print("Enter rows: ");

Int rows = sc.nextInt();

System.out.print("Enter columns: ");

Int cols = sc.nextInt();

Int[][] arr = new int[rows][cols];

Int sum = 0;

```
System.out.println("Enter elements:");
```

```
For (int i = 0; i < rows; i++) {  
    For (int j = 0; j < cols; j++) {  
        Arr[i][j] = sc.nextInt();  
        Sum = sum + arr[i][j];  
    }  
}
```

```
System.out.println("Sum = " + sum);  
}  
}
```

### 3. Find the largest element

```
Import java.util.Scanner;
```

```
Public class Largest2DArray {  
    Public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter rows: ");
```

```
        Int rows = sc.nextInt();
```

```
        System.out.print("Enter columns: ");
```

```
        Int cols = sc.nextInt();
```

```
int[][] arr = new int[rows][cols];
```

```
System.out.println("Enter elements:");
```

```
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < cols; j++) {  
        arr[i][j] = sc.nextInt();  
    }  
}
```

```
int largest = arr[0][0];
```

```
for (int i = 0; i < rows; i++) {  
    for (int j = 0; j < cols; j++) {  
  
        if (arr[i][j] > largest) {  
            largest = arr[i][j];  
        }  
    }  
}
```

```
System.out.println("Largest Element = " + largest);  
}  
}
```

4. Find the sum of each row  
import java.util.Scanner;

```
public class RowSum2DArray {
```

```

Public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter rows: ");
    Int rows = sc.nextInt();

    System.out.print("Enter columns: ");
    Int cols = sc.nextInt();

    Int[][] arr = new int[rows][cols];

    System.out.println("Enter elements:");

    For (int i = 0; i < rows; i++) {
        For (int j = 0; j < cols; j++) {
            Arr[i][j] = sc.nextInt();
        }
    }

    For (int i = 0; i < rows; i++) {

        Int sum = 0;

        For (int j = 0; j < cols; j++) {
            Sum = sum + arr[i][j];
        }

        System.out.println("Sum of Row " + (i + 1) + " = " + sum);
    }
}

```

##### 5. Find the sum of each column

```

Import java.util.Scanner;

```

```

Public class ColumnSum2DArray {

```

```
Public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.print("Enter rows: ");
    Int rows = sc.nextInt();

    System.out.print("Enter columns: ");
    Int cols = sc.nextInt();

    Int[][] arr = new int[rows][cols];

    System.out.println("Enter elements:");

    For (int i = 0; i < rows; i++) {
        For (int j = 0; j < cols; j++) {
            Arr[i][j] = sc.nextInt();
        }
    }

    For (int j = 0; j < cols; j++) {

        Int sum = 0;

        For (int i = 0; i < rows; i++) {
            Sum = sum + arr[i][j];
        }
    }
}
```

```
        System.out.println("Sum of Column " + (j + 1) + " = " + sum);  
    }  
}  
}
```